Lecture 11: Testing Multiple Equations and Review of Inference

Today, we will finish our study of inference by discussing how to test multiple equations. After that, we will do some practice problems covering the different inference topics that we have studied over the last few lectures.

Testing the Effects of Grants

Let us return to the de Mel, McKenzie, and Woodruff (2008) example from last time. They study the effect of different types of grants on the monthly profits of microenterprises in Sri Lanka. We have:

$$\begin{aligned}\text{Profit}_{it} = \alpha + \beta_{LC}\text{Large Cash Grant}_{it} + \beta_{SC}\text{Small Cash Grant}_{it} \\ + \beta_{LIK}\text{Large In-Kind Grant}_{it} + \beta_{SIK}\text{Small In-Kind Grant}_{it} + \gamma_t + \varepsilon_{it}\end{aligned}$$

Last time, we discussed how we might test restrictions that can be expressed as a single equation: for example, we might simplify the regression to just compare cash vs. in-kind grants, and then see if the two grants have the same effect ($\beta_C = \beta_{IK}$), or we might just compare small and large grants, to see if the small grants (which are half the size of the large grants) have half the effect of the large grants ($\beta_S = \frac{1}{2}\beta_L$).

However, we may also want to test hypotheses that consist of multiple equations. For example, instead of simplifying the regression, we may want to use the full regression with four treatments, and compare small cash grants to small in-kind grants, and large cash grants to large in-kind grants. This null hypothesis would consist of two equations:

$$\begin{aligned}\beta_{SC} &= \beta_{SIK} \\ \beta_{LC} &= \beta_{LIK}\end{aligned}$$

Or, we may wish to test the hypothesis that small grants have half the effect of their large counterparts, for cash and separately for in-kind:

$$\begin{aligned}\beta_{SC} &= \frac{1}{2}\cdot\beta_{LC} \\ \beta_{SIK} &= \frac{1}{2}\cdot\beta_{LIK}\end{aligned}$$

Or, we may wish to test the hypothesis that "a rupee is a rupee," meaning that it doesn't matter whether the grant is cash vs. in-kind, and also that large grants just have twice the effect of the small grants:[1]

$$\begin{aligned}\beta_{SC} &= \beta_{SIK} \\ &= \frac{1}{2}\cdot\beta_{LC} \\ &= \frac{1}{2}\cdot\beta_{LIK}\end{aligned}$$

Each of these hypotheses consists of multiple (linear) equations. Note that the null hypothesis is that *all* the equations are true: the hypothesis would be false if any of the equations does not hold.

Today we will cover how to construct valid tests of these types of hypotheses.

[1] Notice that this null hypothesis consists of three equations, since once we have imposed that $\beta_{SC} = \beta_{IK}$, and imposed that the small grants have half the effect of the large grants, it is implied that $\beta_{LC} = \beta_{LIK}$.

The Multiple Testing Problem

Suppose that we are testing the null that $\beta_{SC} = \beta_{SIK}$ and $\beta_{LC} = \beta_{LIK}$. The null hypothesis would be false if $\beta_{SC} \neq \beta_{SIK}$, and/or if $\beta_{LC} \neq \beta_{LIK}$.

A natural (and incorrect) approach to this problem would be to test each equation individually, using the t-statistic method that we developed last lecture. This approach would first test $\beta_{SC} = \beta_{SIK}$ with a 5% test, and then test $\beta_{LC} = \beta_{LIK}$ with a 5% test.

The problem with this approach is that because it tests multiple times, it does not achieve the correct size: it rejects under the null hypothesis too often. Suppose that the null hypothesis was actually true. In that case, a 5% test of $\beta_{SC} = \beta_{SIK}$ will reject 5% of the time, and a 5% test of $\beta_{LC} = \beta_{LIK}$ will also reject 5% of the time. But unless the two tests are perfectly correlated with each other, sometimes one test will reject and the other will not. This means that the probability of *either* test rejecting will be greater than 5%. This problem gets worse the more hypotheses that you test: if we ran 10 independent tests at the 5% level, then we would reject the null $1 - 0.95^{10} \approx 40\%$ of the time.

Note that this is actually a more general problem: in the course of analyzing the data, we often show many coefficients and test many hypotheses. If we have many hypotheses, then some of them are likely to be rejected by chance. This problem becomes even more severe if we only report/publish the results that turn out to be statistically significant: this is called publication bias. One attempt to reduce publication bias has been the rise of pre-analysis plans: for experimental work, economists will now often write down in advance what analysis they will run, and file their plan with a centralized registry. Some academic journals will even accept a paper for publication before the results are known, based solely on the pre-analysis plan. This helps to reduce publication bias, since it ensures that the paper's publication depends on the quality of the research design, but is not conditional on the results of the tests.

The F-Test

A better way to test multiple hypotheses is the F-test. The main idea of the F-test is to compare how well we can fit the data with a "restricted model" that imposes the null hypothesis, versus how well we can fit the data under the unrestricted model. The unrestricted model is simply our standard OLS regression. For comparison, recall that OLS solves the following problem:

$$\min_{\alpha,\beta_1,\ldots,\beta_K} \sum_{i=1}^{N} (Y_i - \alpha - \beta_1 X_{1,i} - \cdots - \beta_K X_{K,i})^2$$

that is, OLS minimizes the sum of squared residuals. We will refer to this sum of squared residuals as SSR_u, where u stands for "unrestricted."

We define the restricted model similarly, but instead of an unconstrained minimization problem, it is a minimization problem where the null hypothesis is imposed as a constraint. For example, suppose our null hypothesis is $\beta_1 = 0$ and $\beta_2 + \beta_3 = 0$. Then, our restricted model would be:

$$\begin{aligned} \min_{\alpha,\beta_1,\ldots,\beta_K} & \sum_{i=1}^{N} (Y_i - \alpha - \beta_1 X_{1,i} - \cdots - \beta_K X_{K,i})^2 \\ & \text{s.t.} \\ \beta_1 &= 0 \\ \beta_2 + \beta_3 &= 0 \end{aligned}$$

and we would use SSR_r to denote the sum of squared residuals from the restricted model. Note that we will always have $SSR_u \leq SSR_r$, since the unrestricted model always fits the data at least as well as the restricted model.

If the null hypothesis is true, then we would expect that the restrictions imposed by the null will not have much effect on the sum of squared residuals, since we are simply imposing a restriction that is actually true. So, if the sum of squared residuals increases a lot when we restrict the model, that is a sign that the null hypothesis is false.

The F-statistic This idea is reflected in the F-statistic, which is our test statistic for the F-test:

$$F = \frac{(SSR_r - SSR_u)/q}{SSR_u/(N-K-1)}$$

where N is the sample size, K is the number of β coefficients (so the total number of coefficients, including the intercept α, is $K+1$), and q is the number of equations being tested. Note that since $SSR_u \leq SSR_r$, the F-statistic is always positive.

This statistic is very intuitive: the term $(SSR_r - SSR_u)/SSR_u$ captures the percentage increase in the sum of squared residuals when we impose the restriction. The larger that percentage increase, the stronger the evidence that the restricted model is incorrect, and the null hypothesis is false. The F-statistic is proportional to $N-K-1$, which represents the sample size, minus a degrees of freedom correction for the number of estimated parameters. Larger samples make us more confident that changes in SSR reflect real differences rather than noise. Finally, the F-statistic is inversely proportional to the number of equations we are testing, q: the more equations we test, the more we expect the SSR to shift (even if the null is true), and so we must adjust the F-statistic to compensate.

Critical Values for the F-Statistic: The F-distribution and the χ^2-distribution

There are two distributions that we can compare the F-test to. These are called the F-distribution and the χ^2-distribution: we can think of them as the multiple-equation analogues of the t-distribution and the normal distribution.

By default, Stata will compare the F-test to the F-distribution, which we denote $F_{q,N-K-1}$. The first number, q, refers to the number of equations being tested, and is called the "numerator degrees of freedom." The second number, $N-K-1$, is the "denominator degrees of freedom." This denominator degrees of freedom is used to make a small-sample adjustment, similar to the one made for the t-distribution.

Much like the t-distribution, the F-distribution converges to a simpler distribution when the denominator degrees of freedom $(N-K-1)$ grows large. As $(N-K-1) \to \infty$, the $F_{q,N-K-1}$ distribution converges to a $\frac{\chi^2_q}{q}$ distribution. In decent-sized samples, the $F_{q,N-K-1}$ distribution and the $\frac{\chi^2_q}{q}$ distribution give nearly identical critical values (just remember to divide the χ^2_q by q).

The χ^2_q distribution is closely related to the standard normal. Suppose that the random variable Z_i is drawn iid, and distributed $N(0,1)$. If we take q draws of Z_i, then $\sum_{i=1}^{q} Z_i^2$ will be distributed χ^2_q.

To test our null hypothesis, we would select the appropriate critical value, c, to ensure that the F-test rejects with the desired probability under the null. We could either use the $F_{q,N-K-1}$ distribution as our reference, or use the χ^2_q distribution and then divide by q. If we wanted to perform a 5% test, then we would select the 95th percentile.

Robust Testing of Multiple Equations

The F-statistic is great for building intuition about what we are doing when we test hypotheses with multiple equations. In essence, we are testing how much worse the model fits when we constraint it to meet those restrictions. It also has a simple formula that is helpful for building intuition.

There is also a huge problem with the F-statistic: it is only valid under homoskedasticity. As we discussed earlier, we generally have no reason to believe that the errors are homoskedastic, and so in general the F-statistic will not give us valid hypothesis tests.

Fortunately, there is a heteroskedasticity-robust version of the F-test. Although the details of that test require matrix algebra and so are beyond the scope of this course, we can understand the intuition through our understanding of the t-statistic, which is asymptotically standard normal. Suppose that for each equation in my null hypothesis, I computed a t-statistic, $t_1, ..., t_q$. If each of those t-statistics were independent, then we could square each t-statistic, and the sum of squared t-statistics would be distributed χ^2_q. If we then divided by q, the test statistic would have our desired distribution. Of course, in reality our t-statistics are typically not independent from each other, but it turns out that there is a way to use the covariance matrix

to "rotate" the t-statistics to make them independent from each other (that is the bit that requires matrix algebra). If we do this with the homoskedastic estimate of the variance matrix, then we get the traditional F-test. If we instead do it with the robust estimator, then we get the robust F-test.

In practice, you should always use the robust version of the F-test, rather than the traditional F-test.

It is easy to implement the robust F-test on the computer in Stata. Suppose that you wish to run the regression

$$Y_i = \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \varepsilon_i$$

and then test the joint hypothesis that $\beta_1 = 0$ and $\beta_2 + \beta_3 = 0$. The best way to do this is with the `test` command. You would run:

```
reg y x1 x2 x3 x4, rob
test (x1 = 0) (x2 + x3 = 0)
```

which will output the robust version of the F-statistic, as well as the associated p-value. Make sure to specify "`, rob`" after the regression, otherwise the `test` command will perform the non-robust version of the F-test.

Suppose we are studying the effect of a training program on income. Participants are randomly assigned to one of three groups: control (no training), basic training, or advanced training. We estimate the following regression:

$$\text{Monthly Income}_i = \alpha + \beta_1 \text{Basic}_i + \beta_2 \text{Advanced}_i + \varepsilon_i$$

where Basic_i is a dummy variable equal to one if the individual was assigned to basic training, and Advanced_i is a dummy variable equal to one if the individual was assigned to advanced training.

Exercise 1. Which group is the omitted category? What difference in means does β_1 represent? What difference in means does β_2 represent?

Exercise 2. Suppose I want to test whether basic training affects income. What is my null hypothesis? How would I construct a 5% test of this null? Set up the t-statistic and state the appropriate critical value.

Exercise 3. Suppose I want to test whether basic training and advanced training have the same effect. What is my null hypothesis? How would I construct a 5% test of this null, using the linear combination approach that we outlined in class? How would I implement that approach in Stata?

Exercise 4. Suppose I want to test whether basic training and advanced training have the same effect, but I do not want to use the linear combination approach. How could I change the dummy variables in my regression so that one of the coefficients reflects the difference between the basic and advanced training? *Hint: There are actually multiple ways to do this! One approach is to change the omitted category, while another is to convert one of the dummy variables into a generic "Training" dummy. Explain why those approaches work, using your understanding of dummy variables.*

Exercise 5. Returning to the original form of the regression, suppose that I want to test the hypothesis that neither training has any effect. State that null hypothesis. How would I test that null? How would I implement that test in Stata?

Exercise 6. Suppose that I get the estimates $\hat{\beta}_1 = 60$, $\widehat{\text{SE}}\left(\hat{\beta}_1\right) = 50$, $\hat{\beta}_2 = 90$, and $\widehat{\text{SE}}\left(\hat{\beta}_2\right) = 40$. Can we reject, at the 5% level, that basic training has no effect on monthly income? What about advanced training? Can we conclude that basic training has no effect?

Exercise 7. Under the above estimates, what can I say about the effects of basic versus advanced training? Can I use the information above to test whether advanced training is more effective than basic training, at the 5% level?